CSE558 · DATA SCIENCE — LECTURE 7

The Two-Sample Family

Format

Six tests taught in turn, a predict-before-reveal example, then two chi-square previews

Builds on

Last session — the master pattern and χ^2 for a variance

Leads to

Lecture 9 — why these previews work, then MLE

RECAPE

Two lines from last time

The master pattern

(your estimate − the value H_0 claims) ÷ (the standard error of your estimate).

χ^2 for a variance

A ratio, not a difference over an error — and asymmetric critical values, unlike every z or t statistic so far.

A DIFFERENT QUESTION

From one sample to two

Until now

One sample, weighed against a status quo value. The null names a number.

From here on

Two samples, weighed against each other.

Six tests

Two means, two proportions, two means with σ unknown, the paired version of that, a correlation, and two variances.

TEST 1

z-test for two population means

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Use it when

- Both population standard deviations are known.
- Large, roughly normal samples.

$H_0: \mu_1 = \mu_2$, $H_1: \mu_1 \neq \mu_2$

$\bar{X}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} x_i$

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

Does the drug do anything at all at α=0.05?

Group 1 — the drug

$n_1 = 1000$

$\bar{x}_1 = 10$

$\sigma_1 = 4$

Group 2 — the placebo

$n_2 = 900$

$\bar{x}_2 = 9$

$\sigma_2 = 5$

$\bar{X}_1 = \frac{(\quad)}{1000} = 10$

H_1

$\frac{(10-9)-(0)}{\sqrt{\frac{16}{1000}+\frac{25}{900}}} = 4$ ~~8~~ xx

$H_0\ (\mu_1 = \mu_2) \Rightarrow \mu_1 \leq \mu_2$

$H_0\ \mu_1 \leq \mu_2 + 20$ H_1: $\mu_1 \neq \mu_2$

$H_1\ \mu_1 > \mu_2 + 20$ $\mu_1 > \mu_2$

TEST 2

z-test for two proportions

$$Z = \frac{(p_1 - p_2) - (\mu_1 - \mu_2)}{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

The pooled rate

$p = (p_1 n_1 + p_2 n_2)/(n_1 + n_2)$

One combined estimate of the shared success rate.

Why pool?

Only when both samples came from the same population!!

P_1 rate of success in Sample 1

$P_2 \Rightarrow$ Sample 2

WORKED EXAMPLE — Z-TEST FOR TWO PROPORTIONS

$$Z = \frac{p_1 - p_2}{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

Does campaign A converts faster?

$$p = (p_1 n_1 + p_2 n_2)/(n_1 + n_2)$$

Campaign A

$n_1 = 350$

Conversions: 63

Campaign B

$n_2 = 400$

Conversions: 52

H_0 $M_1 \leq M_2$

H_1 $M_1 > M_2$

Independent or paired? It changes the test

Unpaired — two separate groups

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s\sqrt{1/n_1 + 1/n_2}}$$

Pooled $s^2 = \left[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2\right]/(n_1 + n_2 - 2)$

Degrees of freedom = ?

Paired — the same subjects, twice

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Work with the differences d_i only.

$$s_d^2 = \frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2$$

Degrees of freedom = ?

EXAMPLE — T-TEST

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s\sqrt{1/n_1 + 1/n_2}} \qquad t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Twelve patients, before and after

The design

A medical researcher wants to know if the new medication is effective at reducing blood pressure. At the 0.05 level of significance, does the sample data provide sufficient evidence to conclude that the new drug significantly lowers systolic blood pressure?

Metric	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
Before	150	142	160	135	148	155	138	145	152	162	140	150
After	135	138	144	130	142	130	140	135	145	150	135	145

d.f. = 11, critical region 2.209

$d_1 \rightarrow$ 15 4 16 5 6 15 −2 10 7 12 5 5

H_0 $d \leq 0$

H_1 $d > 0$

Correlation, and comparing two spreads

t-test for a correlation coefficient

$$t = \frac{r\sqrt{n-2}}{\sqrt{(1-r^2)}}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Degrees of freedom = n − 2.

Limitation: it only ever sees linear structure.

F-test for two population variances

$$F = \frac{s_1^2}{s_2^2}$$

H_0: $\sigma_1^2 = \sigma_2^2$ — the two spreads match.

Degrees of freedom = (n_1−1, n_2−1).

The two means need not be equal.

d.f. = 25, critical region 2.06

Do study hours predict exam scores?

n = 27 students. Sample correlation between studying and exam score: r = 0.6. H_0: r = 0 at α = 0.05.

H_1 $r \neq 0$

$$\frac{0.6\sqrt{5}}{\sqrt{1-0.36}} = \frac{0.6\times 5}{0.8}$$

PREDICT BEFORE THE REVEAL

Critical Region = 2.093
Critical Region = 2.080

Does liver fat % differ between men and women?

The data, exactly as recorded

Men (n = 13): 13.3, −6, 20, 8, 14, 19, 18, 25, 16, 24, 15, 1, 105

Women (n = 10): 22, 16, 21.7, 21, 30, 26, 12, 23.2, 28, 23

$\bar{X}_1$ 20 94

H_0 $\mu_1 = \mu_2$

H_1 $\mu_1 \neq \mu_2$

22.29

$S_1^2 = 713.52$

$S_2^2 = 2832$

S_2^2 4198 5

$\frac{-135}{20\sqrt{\left(\frac{1}{13}+\frac{1}{10}\right)}} = 2.0156$

$\frac{(\bar{X}_1 - \bar{X}_2)}{S\sqrt{1/n_1 + 1/n_2}}$

$S^2 = \frac{(n_1-1)S_1^2 + n_2-1S_2^2}{n_1+n_2-2}$

STUDY MATERIAL

Reference

The Art of Learning from Data: Chapter 10.1, 10.2, 10.4, 12.2 by Agresti, Franklin & Klingenberg

Slides